NanoClaw vs OpenClaw — Architecture Comparison

NanoClaw vs OpenClaw — Deep Architecture Comparison

> Diagrams use Mermaid — paste any block into mermaid.live to render interactively.

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1. Top-Level Philosophy

Dimension	NanoClaw	OpenClaw
Core premise	Agent is dangerous, must be caged	Agent is powerful, must be orchestrated
Codebase size	~4,000 lines (intentional)	Enterprise monorepo, 100k+ lines
Security model	OS-level / hardware isolation	Application-layer + operator trust
Extensibility	Static code transformation	Runtime plugin registry (ClawHub)
Deployment target	Single-user, security-critical	Teams, cloud, Kubernetes
Stars / adoption	Niche, growing	215,000+ stars
License	—	MIT

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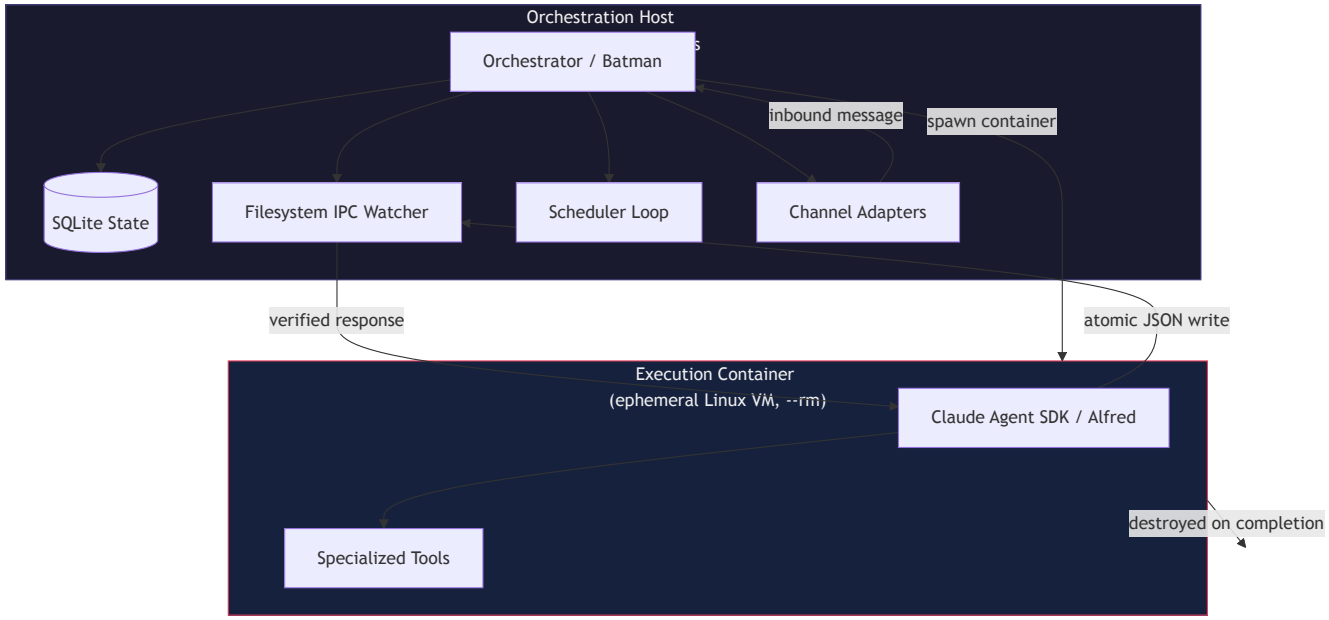
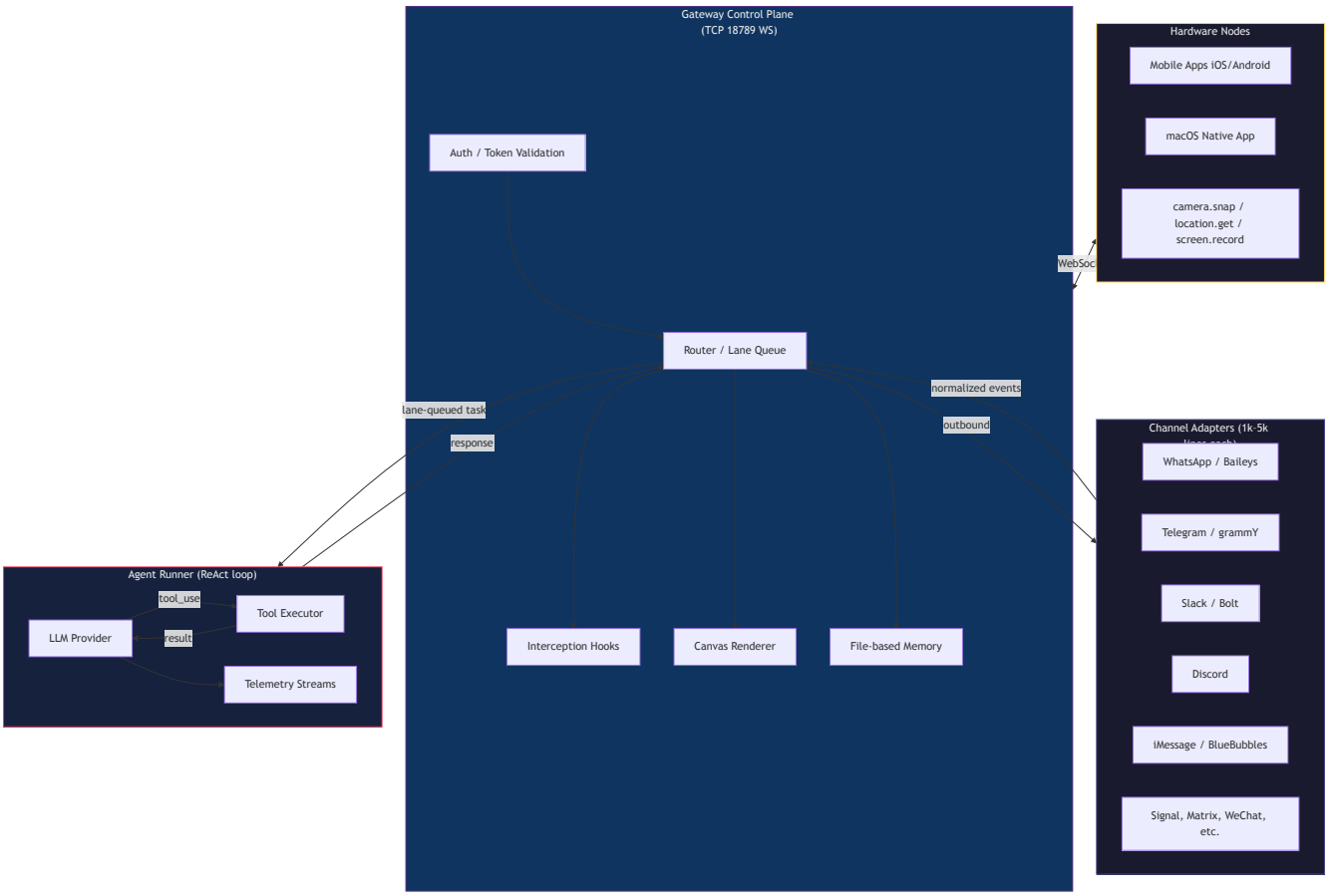
2. High-Level Architecture

NanoClaw

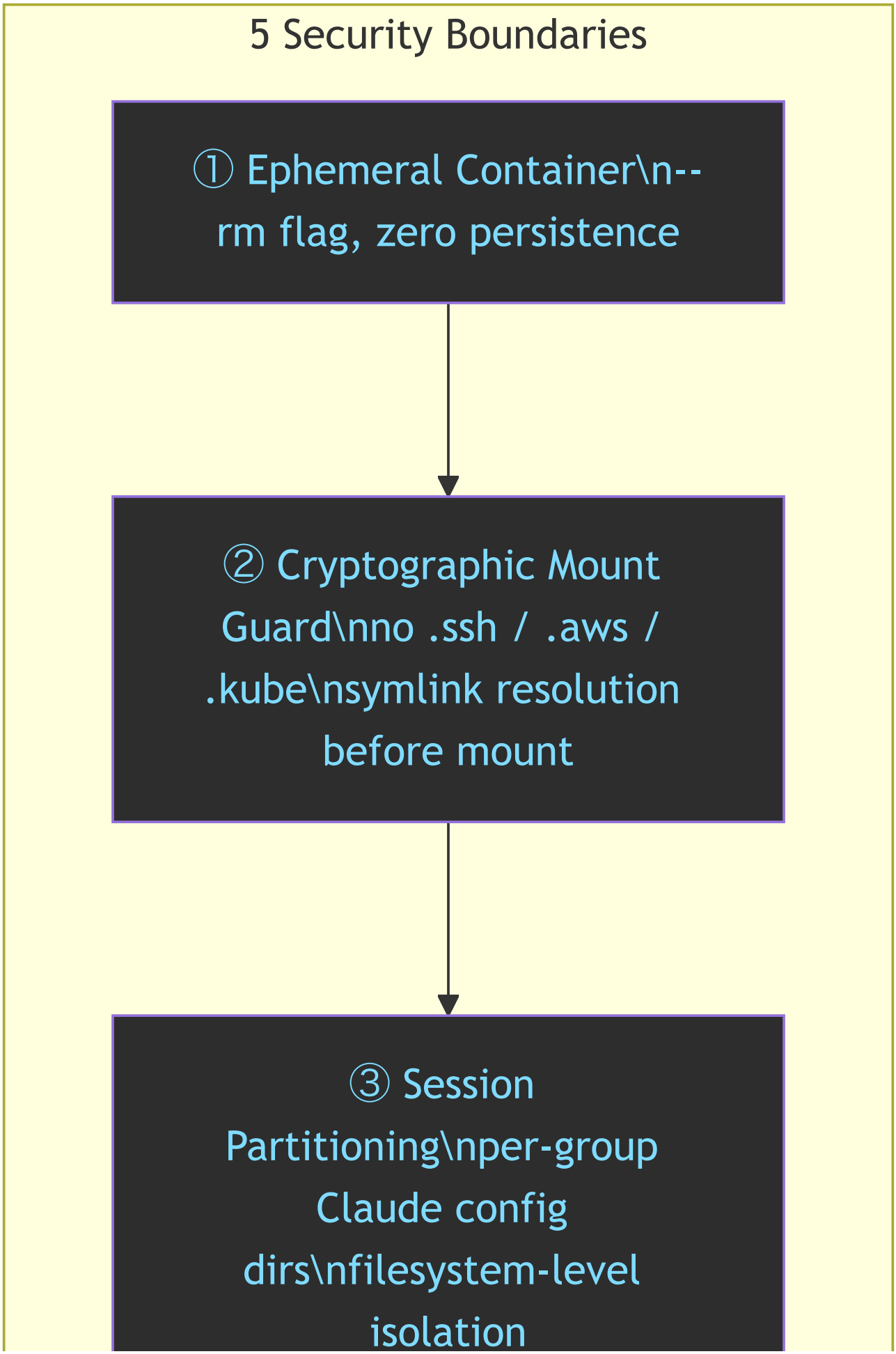


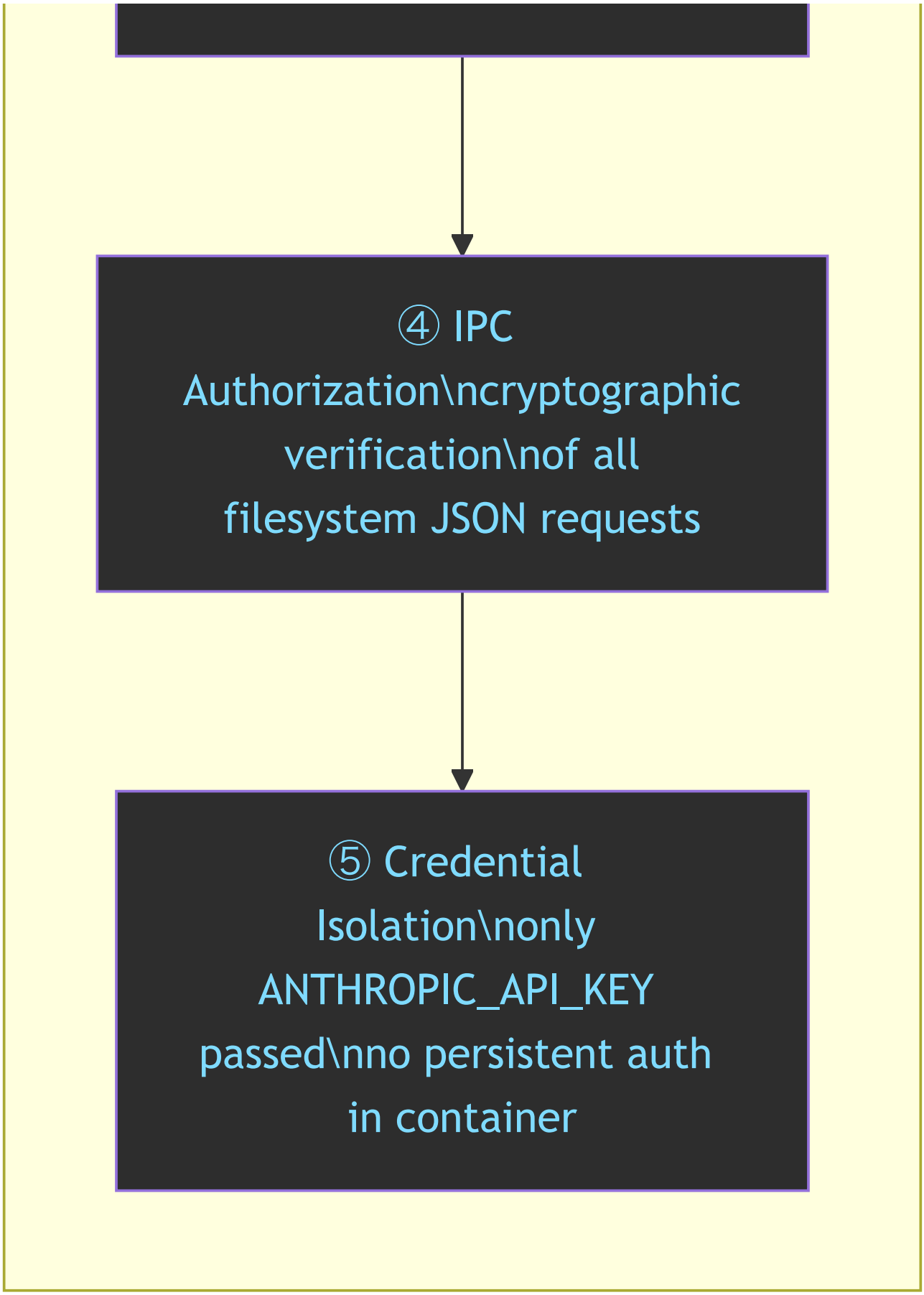
OpenClaw


Gateway Control Plane
(TCP 18789 WS)
Auth / Token Validation
Router / Lane Queue
lane-queued task
response
normalized events
outbound
Interception Hooks
Canvas Renderer
File-based Memory
Hardware Nodes
Mobile Apps iOS/Android
macOS Native App
camera.snap /
location.get /
screen.record
WebSoc
Channel Adapters (1k-5k
WhatsApp / Baileys
Telegram / grammY
Slack / Bolt
Discord
iMessage / BlueBubbles
Signal, Matrix, WeChat,
etc.
Agent Runner (ReAct loop)
tool_use
Tool Executor
LLM Provider
result
Telemetry Streams

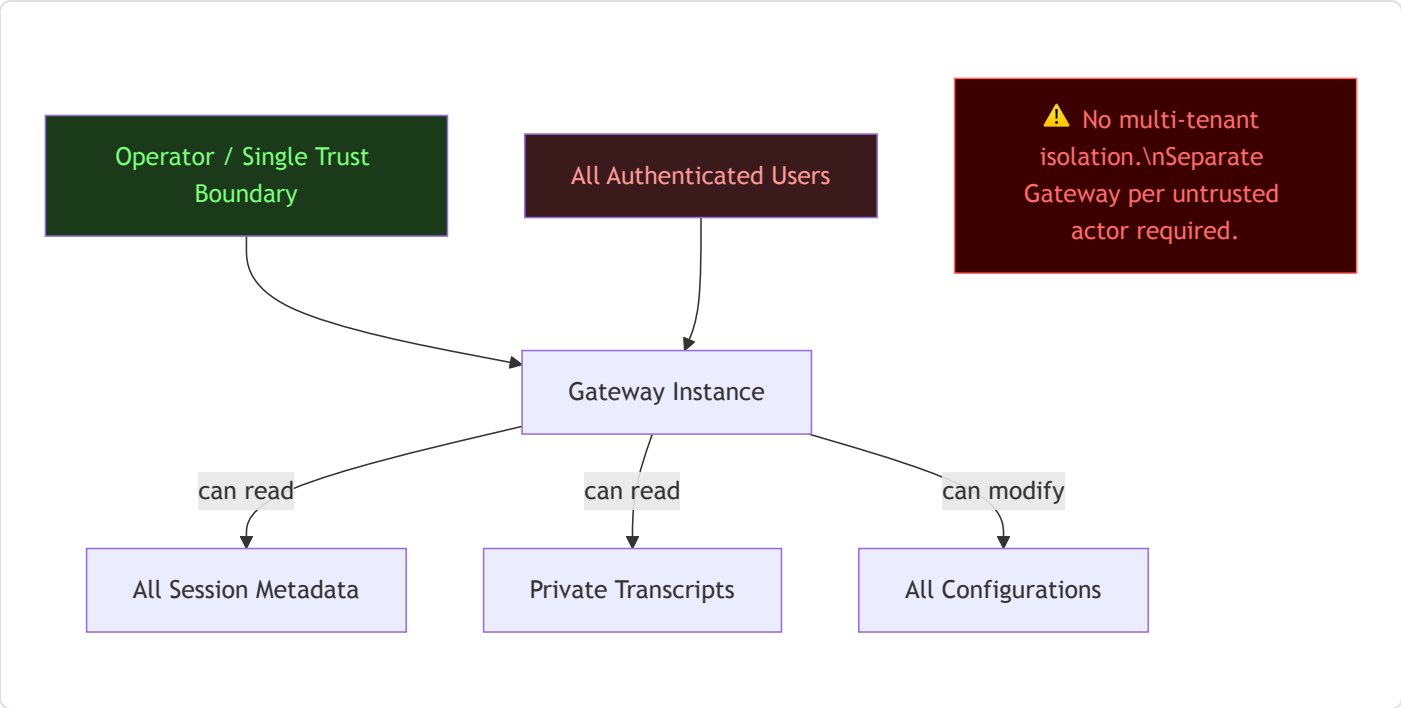
3. Security Architecture

NanoClaw — Defense in Depth (OS-level)


5 Security Boundaries
① Ephemeral Container\n--rm flag, zero persistence
② Cryptographic Mount Guard\nno .ssh / .aws / .kube\nsymlink resolution before mount
③ Session Partitioning\nper-group Claude config dirs\nfilesystem-level isolation


④ IPC Authorization\ncryptographic verification\nof all filesystem JSON requests
⑤ Credential Isolation\nonly ANTHROPIC_API_KEY passed\nno persistent auth in container

OpenClaw — Application-Layer Trust Model



Vulnerability Comparison

Attack Surface	NanoClaw	OpenClaw
Prompt injection → host escape	Blocked by VM boundary	Application-layer mitigations only
Cross-session data leak	Blocked by filesystem partitioning	Possible — single trust boundary
Credential exposure in container	Only API key, no persistent auth	Configurable, defaults vary
Network attack surface	None — containers have no network stack	TCP/18789 WebSocket, loopback default
Supply chain (skills)	Static merge, no runtime exec of skill code	7.1% of ClawHub skills mishandle secrets; 283–341 confirmed credential stealers
Multi-tenant	Not supported (by design — single user)	Not supported (explicit doc disclaimer)
Memory accumulation (CWE-770)	N/A	Known bug in `command-queue.ts` — LaneState Map never cleaned, causes OOM
WebSocket hijacking	N/A	Session fixation via CSRF → persistent bidirectional access

Attack Surface	NanoClaw	OpenClaw
Network misconfiguration	N/A	0.0.0.0 binding → RCE within minutes of public exposure

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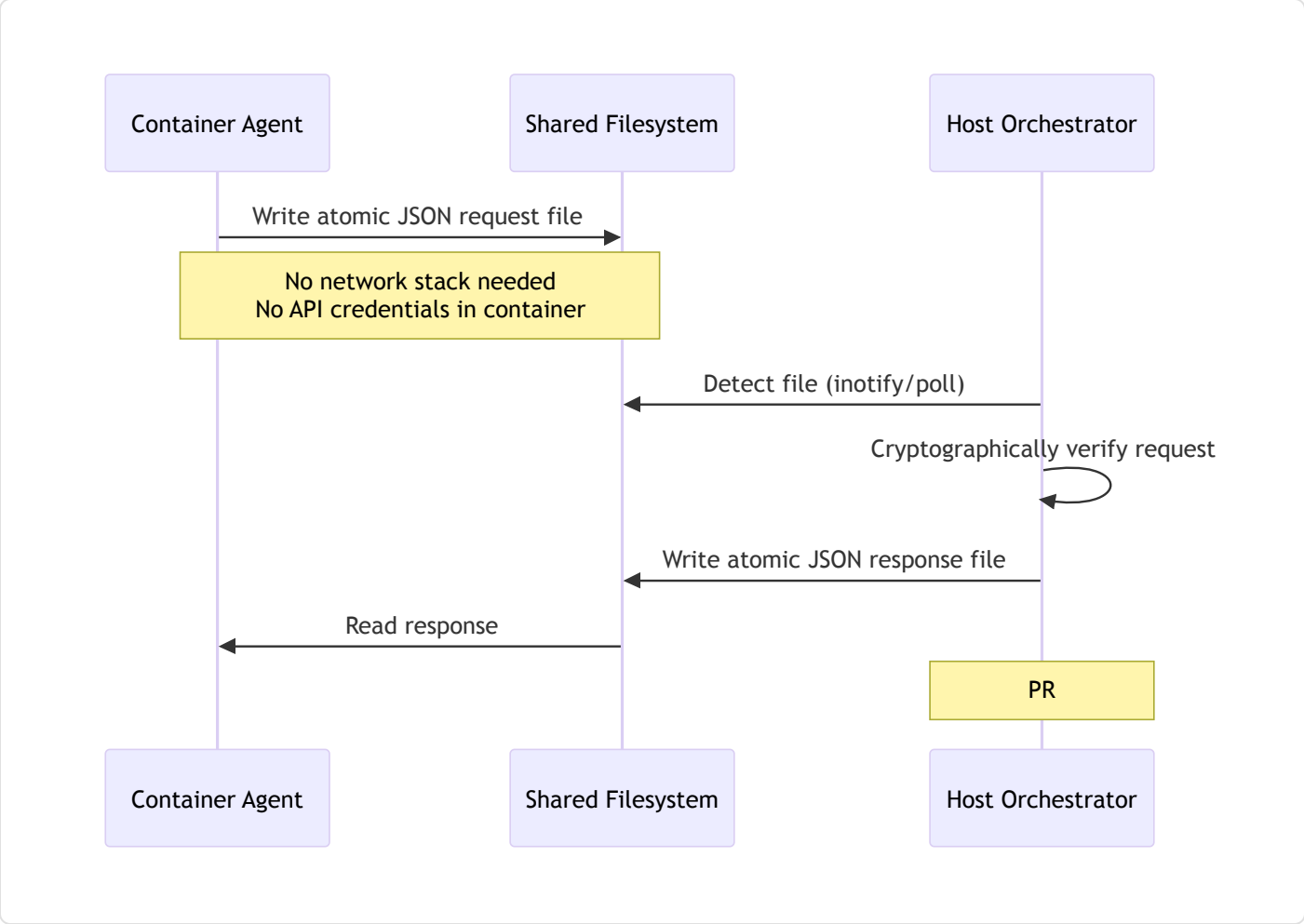
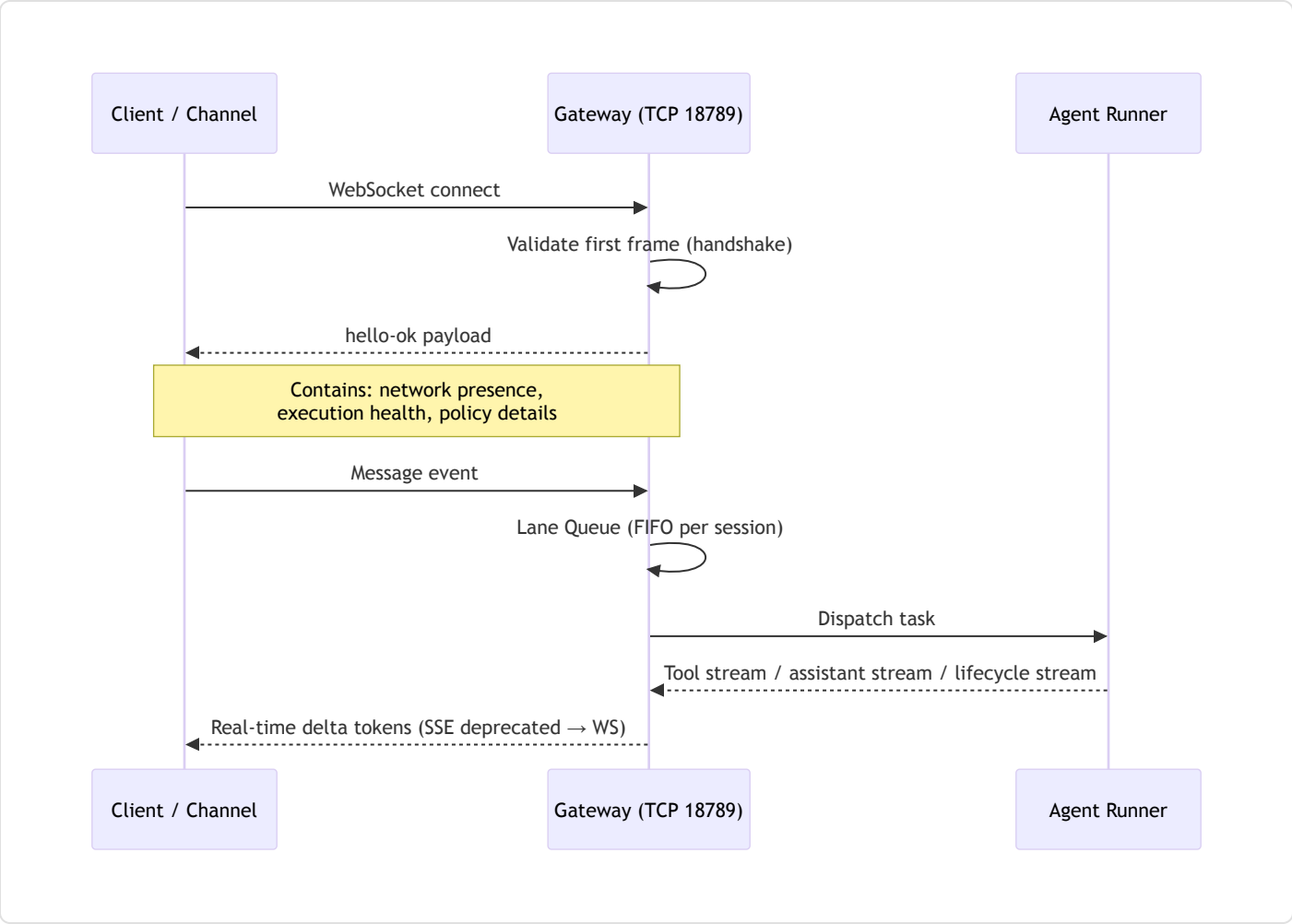
4. Inter-Process Communication

NanoClaw — Filesystem IPC



Properties: - Zero network exposure inside container - Atomic writes prevent stream injection - Request forgery blocked by crypto verification - Host never writes back without verification

OpenClaw — WebSocket Transport



Properties: - Bidirectional, real-time - Strict FIFO per lane prevents race conditions - Unencrypted by default — TLS is operator responsibility - Session fixation risk on hijack

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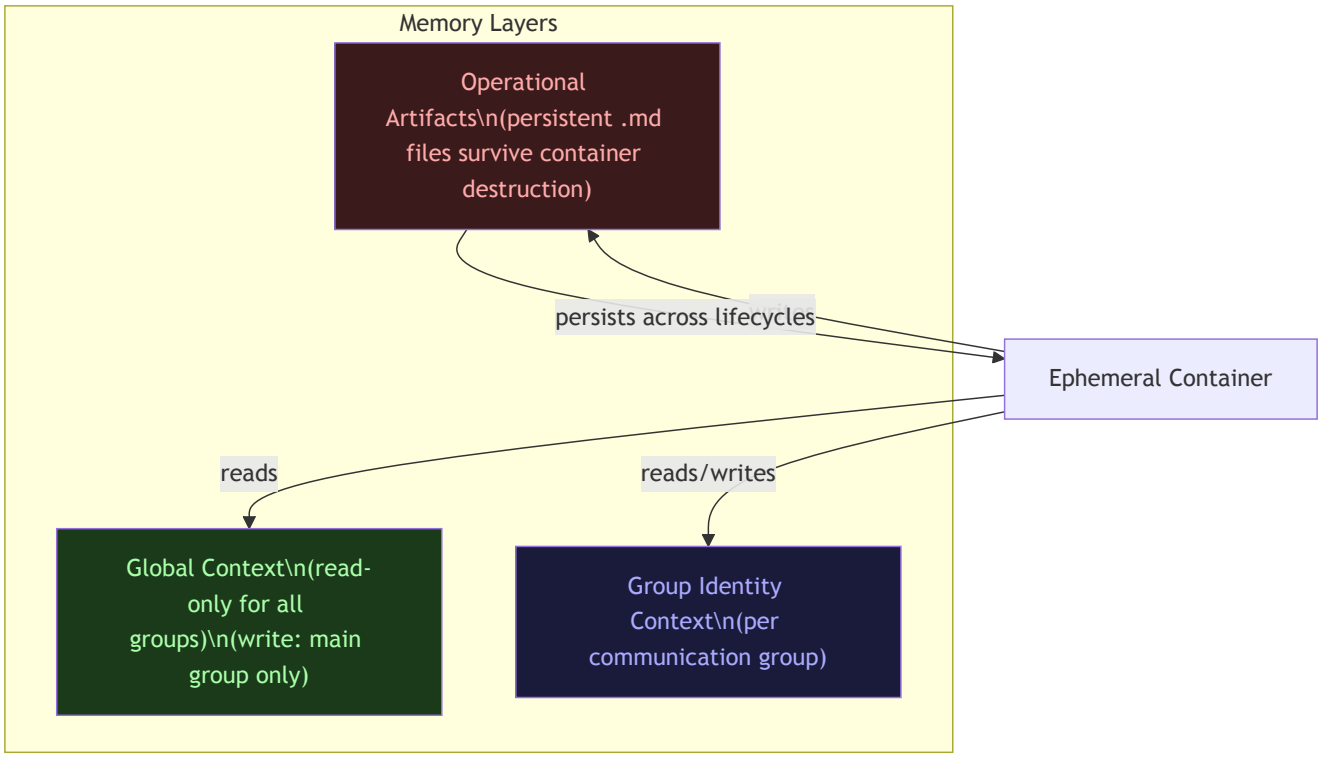
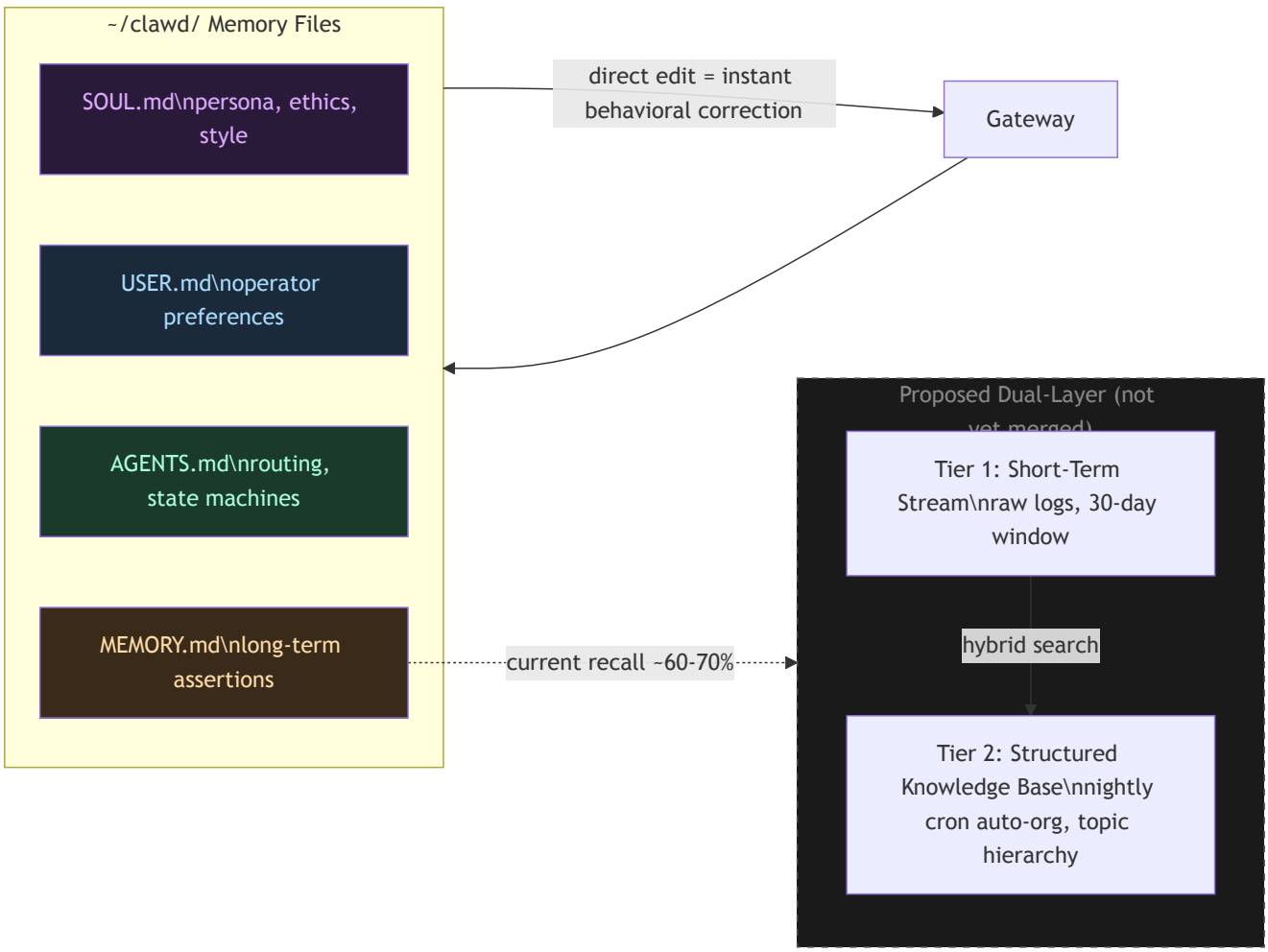
5. Memory Architecture

NanoClaw — Three-Layer Memory

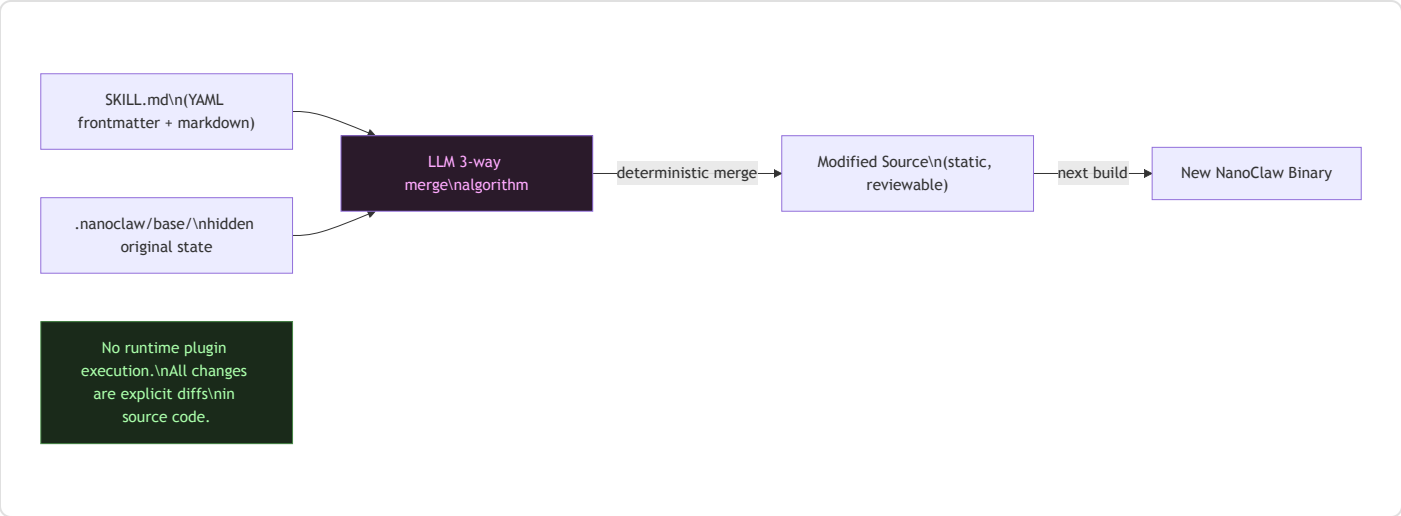


OpenClaw — File-Based Markdown Memory



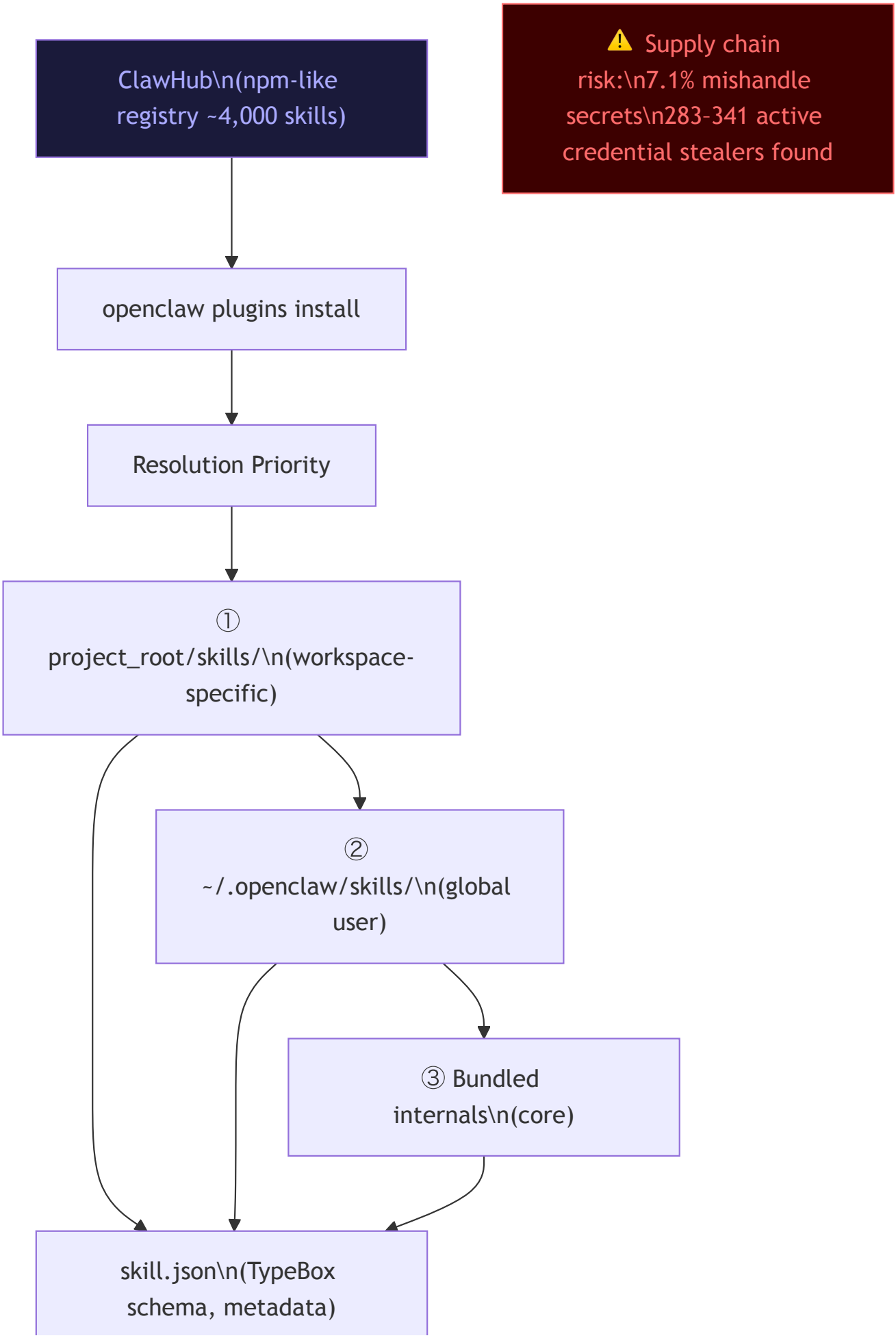
6. Extensibility / Skills Engine

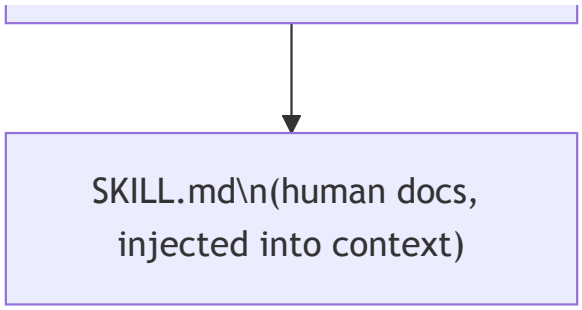
NanoClaw — Static Code Transformation



Key properties: - Skills are merged *into* the codebase, not loaded at runtime - Every integration is an auditable source diff - Codebase stays small (skills don't accumulate as runtime deps) - Follows Anthropic Agent Skills open standard

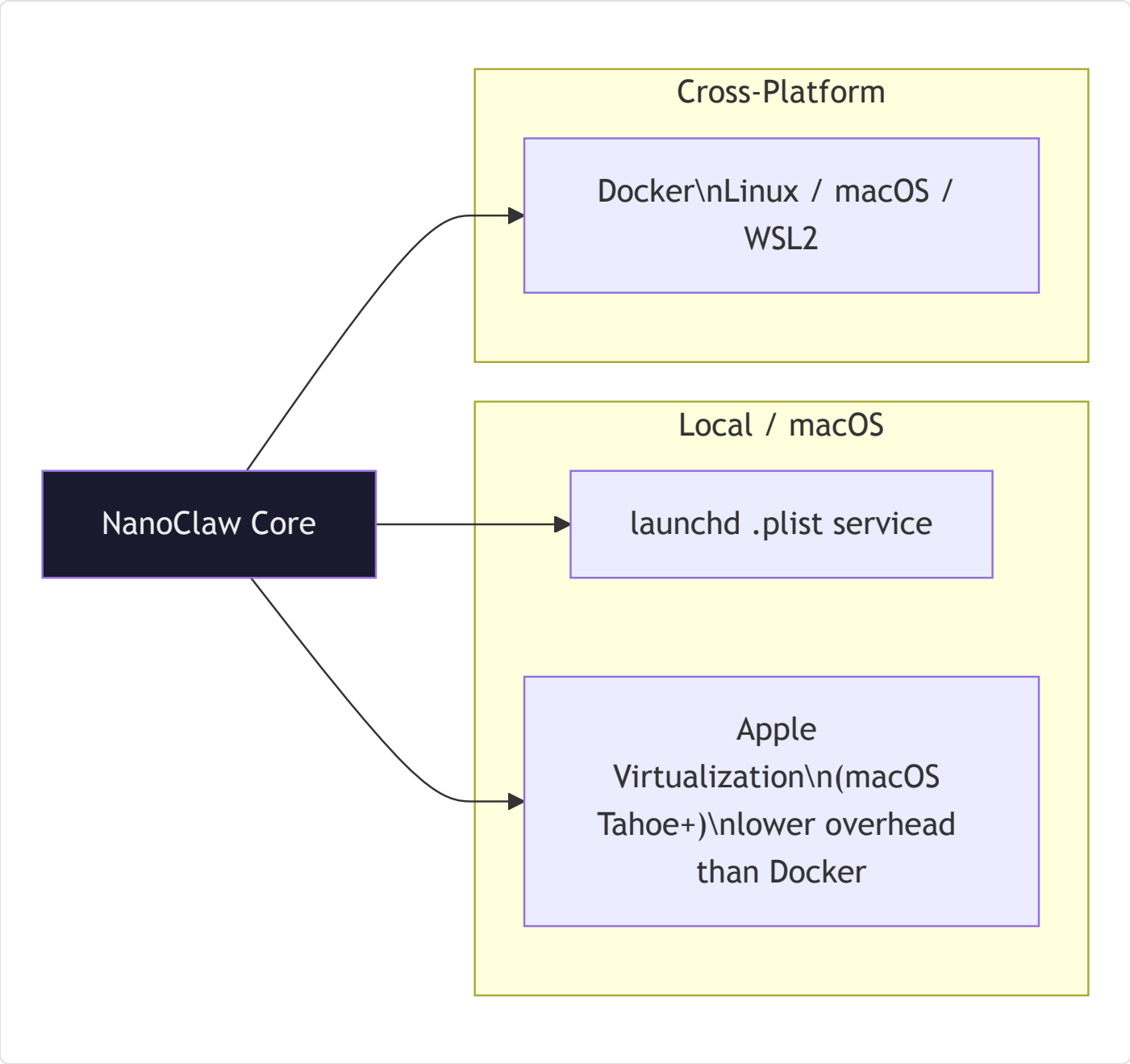
OpenClaw — Runtime Plugin Registry (ClawHub)


ClawHub\n(npm-like registry ~4,000 skills)
⚠️ Supply chain risk:\n7.1% mishandle secrets\n283–341 active credential stealers found
openclaw plugins install
Resolution Priority
① project_root/skills/\n(workspace-specific)
② ~/.openclaw/skills/\n(global user)
③ Bundled internals\n(core)
skill.json\n(TypeBox schema, metadata)

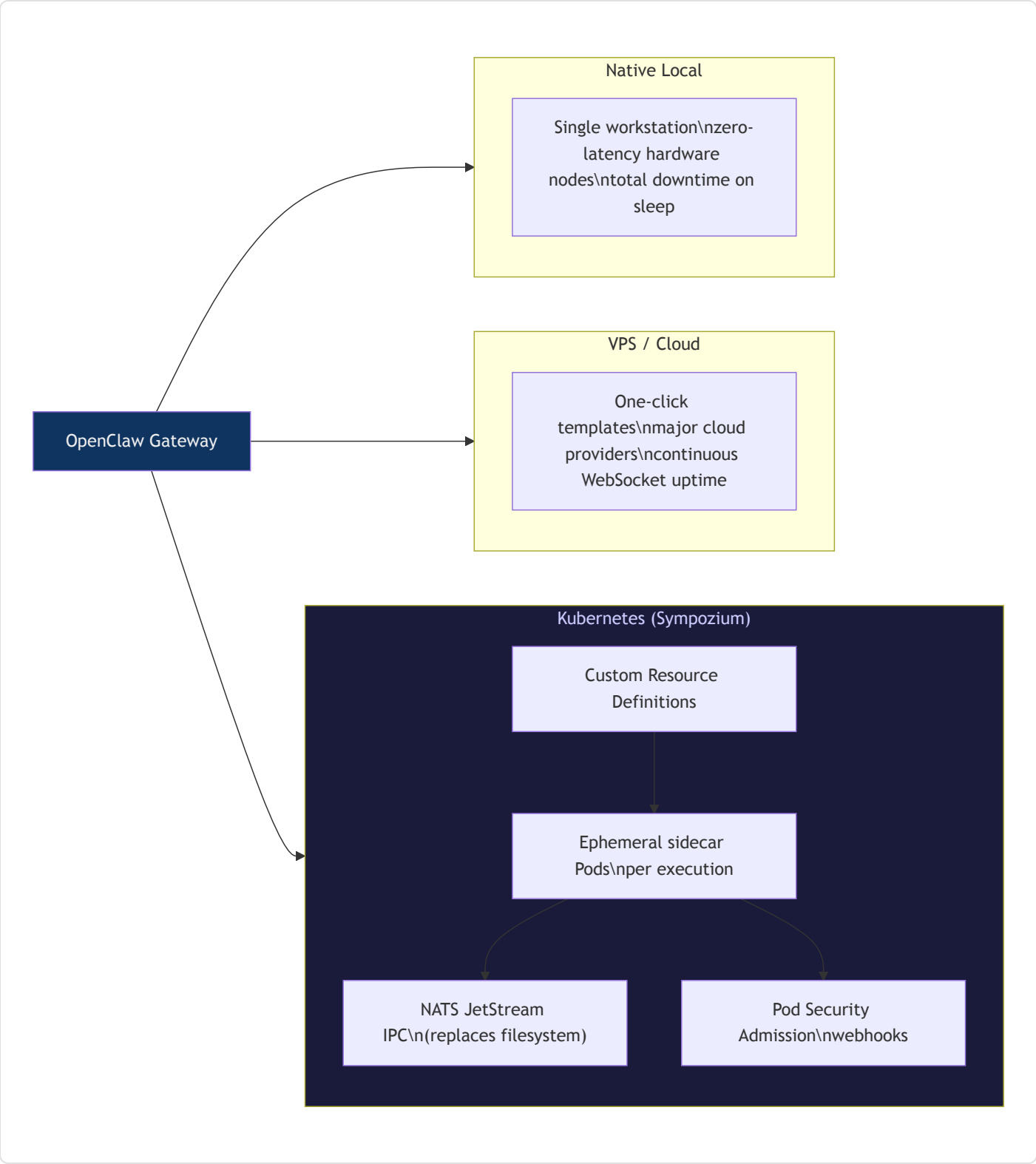

SKILL.md\n(human docs,
injected into context)

7. Deployment Models

NanoClaw



OpenClaw


Native Local
Single workstation\nzero-latency hardware nodes\ntotal downtime on sleep
VPS / Cloud
One-click templates\nmajor cloud providers\ncontinuous WebSocket uptime
OpenClaw Gateway
Kubernetes (Sympozium)
Custom Resource Definitions
Ephemeral sidecar Pods\nper execution
NATS JetStream IPC\n(replaces filesystem)
Pod Security Admission\nwebhooks

8. Channel Support (Current + In-Flight PRs)

Channel	NanoClaw	OpenClaw
Telegram	✅ core	✅ core
Slack	✅ core	✅ core
Discord	✅ skill	✅ extension
WhatsApp	✅ skill	✅ Baileys
Signal	✅ skill (PRs #784, #665)	✅ extension
iMessage	—	✅ BlueBubbles
Feishu / Lark	✅ skills (3 PRs)	✅ heavy (6+ PRs)
Matrix	✅ PR #791	—
Mattermost	✅ PR #546	✅ extension
Google Chat	✅ PR #752	—
QQ / NapCat	✅ PRs #821, #796	—
WeChat Work	—	✅ PR #39511
DingTalk	✅ PR #764	—
Web Chat (browser)	✅ PR #797	✅ core web-ui
IMAP Email	—	✅ PR #39625
CLI	✅ PR #680	✅ core
Hardware nodes (camera/location)	❌	✅ core

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9. AI Provider Support

Provider	NanoClaw	OpenClaw
Anthropic Claude	✅ primary	✅
Ollama (local)	✅ PR #797	✅
llama.cpp	✅ PR #762	—
Generic LLM / OpenAI-compat	✅ PR #557	✅
Azure OpenAI	—	✅ PR #39540

Provider	NanoClaw	OpenClaw
Codex	✅ PR #572	—
Novita AI	—	✅ PR #39675
GitHub Copilot models	—	✅ PR #39613
Moonshot / Kimi	—	✅

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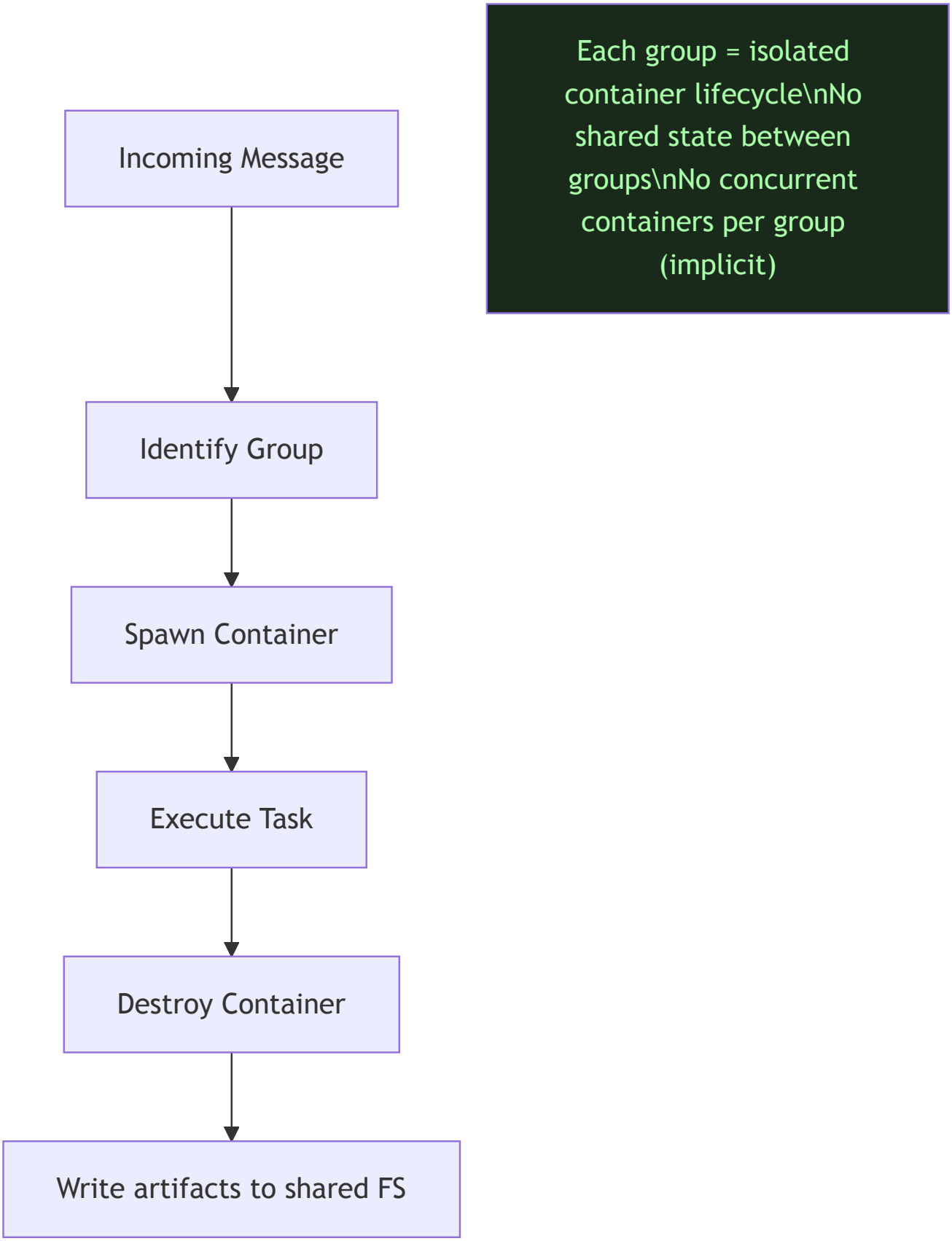
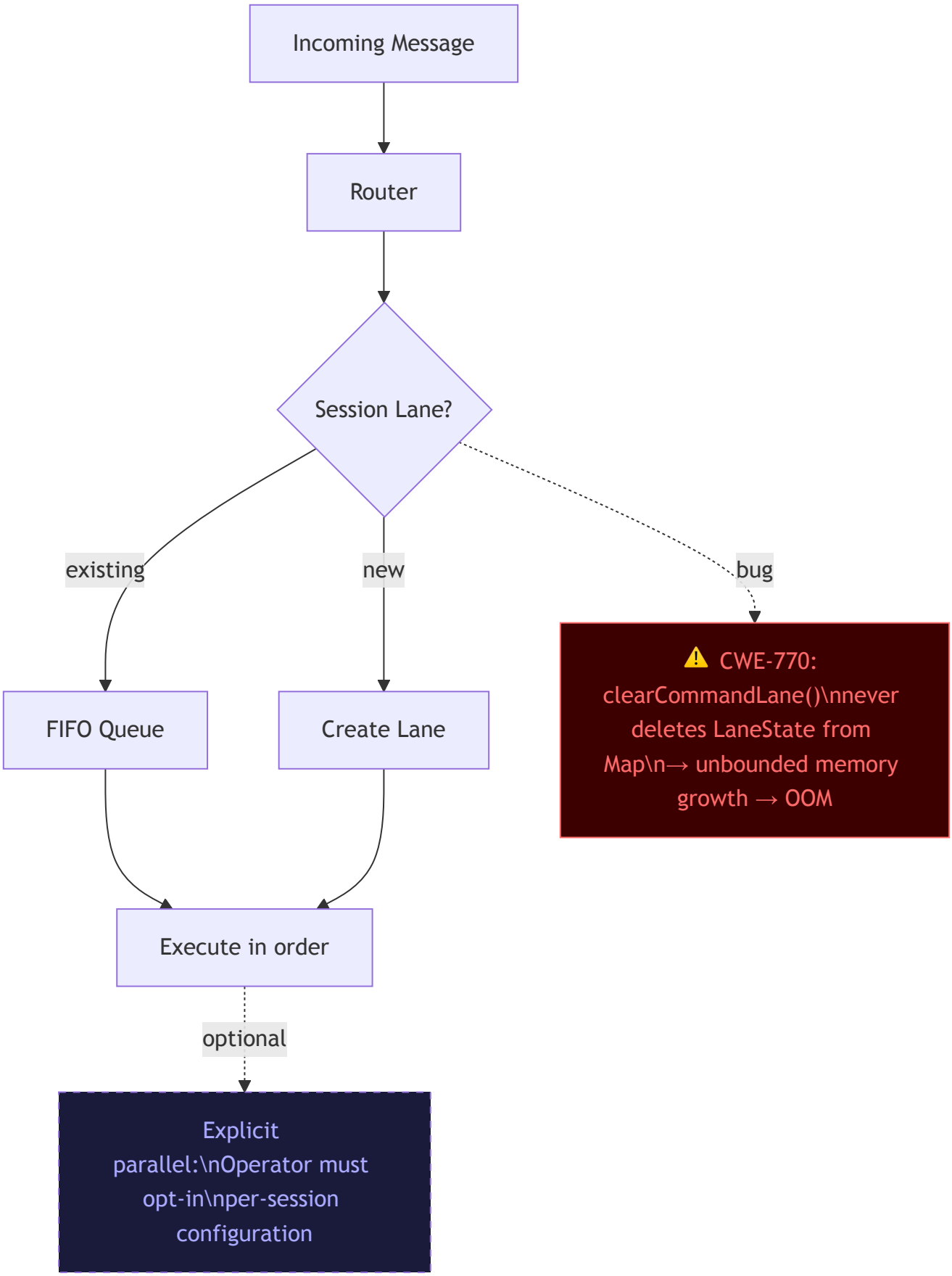
10. Concurrency Model

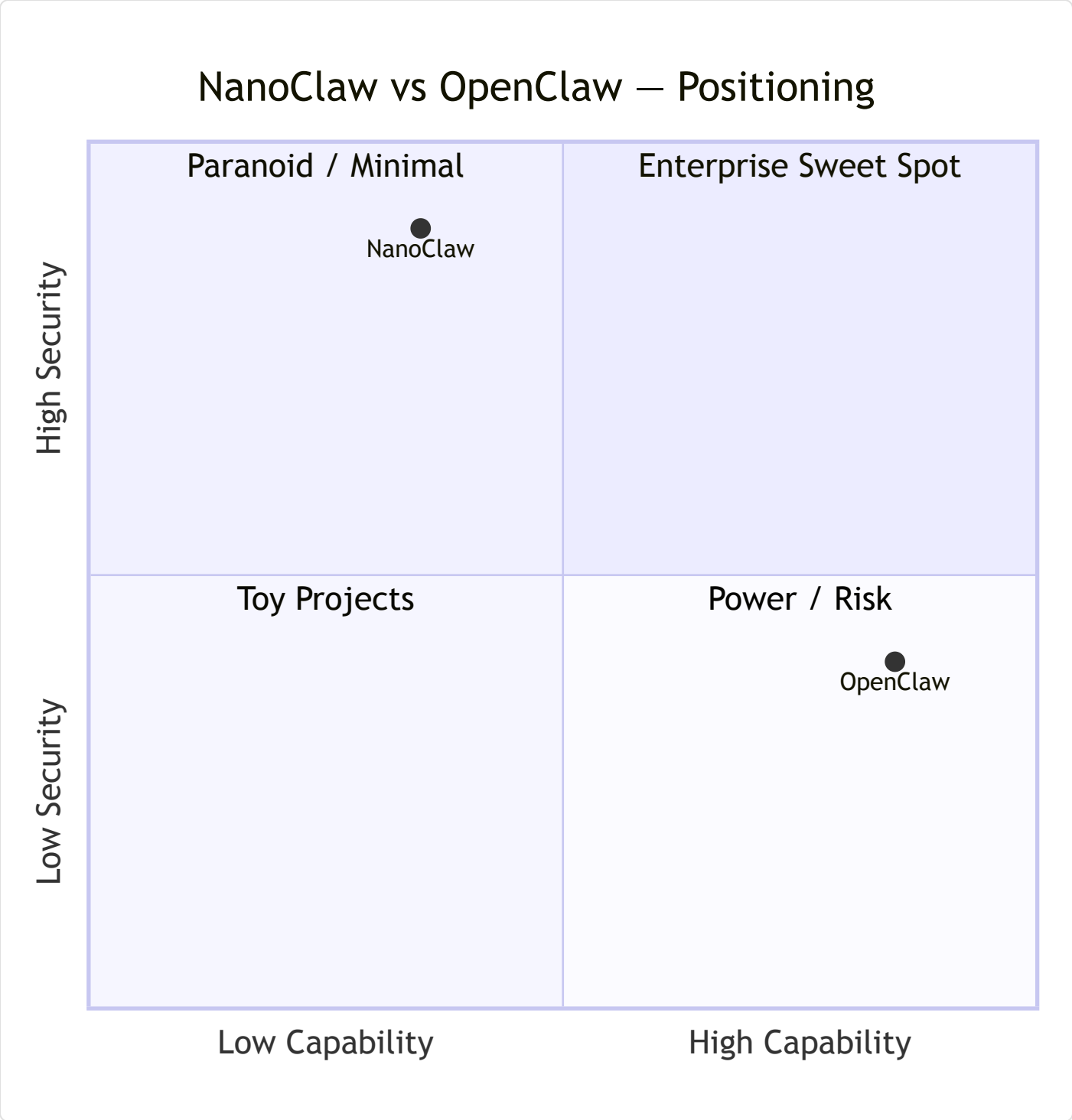
NanoClaw


Incoming Message
Identify Group
Spawn Container
Execute Task
Destroy Container
Write artifacts to shared FS
Each group = isolated container lifecycle\nNo shared state between groups\nNo concurrent containers per group (implicit)

OpenClaw — Lane Queue (Default Serial, Explicit Parallel)


Incoming Message
Router
Session Lane?
existing
new
bug
FIFO Queue
Create Lane
⚠️ CWE-770: clearCommandLane()\nnever deletes LaneState from Map\n→ unbounded memory growth → OOM
Execute in order
optional
Explicit parallel:\nOperator must opt-in\nper-session configuration

11. What's Actually Different — Summary Matrix



Concern	NanoClaw wins	OpenClaw wins
Security isolation	✅ OS-level, provable	
Attack surface	✅ Minimal, auditable	
Channel breadth		✅ More native integrations
Hardware node support		✅ Camera, location, screen

Concern	NanoClaw wins	OpenClaw wins
Kubernetes / cloud scale		✅ Sympozium
AI provider variety		✅ More providers
Community size		✅ 215k stars
Codebase auditability	✅ 4k lines, LLM-readable	
Skill supply chain safety	✅ Static merge, no runtime exec	
Memory debuggability	✅ Filesystem + SQLite	✅ Markdown edit = instant fix
Real-time streaming	Filesystem-based (lower latency risk)	✅ WebSocket, SSE
Mobile apps	❌	✅ iOS + Android
Canvas / visual workspace	❌	✅ HTML/CSS/JS agent-driven
Multi-tenant safety	❌ (both fail, NanoClaw by design)	❌ (explicit disclaimer)

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12. Active Development Signals (from open PRs)

NanoClaw (open PR analysis)

- **IPC rewrite** (#816) — moving to JSON-RPC 2.0 over stdio (architectural shift away from filesystem IPC)
- **Web Chat UI** (#797) — 77k line PR adding browser channel, Cloudflare tunnel, Ollama, Airtable
- **Memory system** (#560/#561) — RAG-based semantic memory being added
- **New channels flood** — QQ, WeChat Work via NapCat, Matrix, Mattermost, Google Chat

OpenClaw (open PR analysis)

- **Feishu mega-PR** (#39496) — 9 streaming bug fixes, 3-layer dedup, calendar CRUD tools
- **IMAP hook** (#39625) — email without Gmail dependency
- **Azure OpenAI + Novita** — provider expansion
- **Wecom channel** (#39511) — WeChat Work already on npm, requesting inclusion
- **Per-agent timezone** (#39610) — cross-cutting config quality-of-life
- **Dual-layer memory** — proposed upgrade from ~65% to higher recall (not yet in PR)

> **Bottom line:** NanoClaw is being pushed toward richer features (web UI, RAG memory, more channels) while trying to maintain its security-first identity. OpenClaw is maturing its core (fixing IPC bugs, better memory, provider breadth). The gap in security model is structural — NanoClaw's container boundary is a design constraint that OpenClaw doesn't share and can't easily retrofit.